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Analyzing and Planning Stock for Q-Commerce



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Demand Forecasting for August 2022

Business Context

Background

To heighten company efficiency and increase customer satisfaction, the operation team needs to know the forecast demand for the next 1 month (August 2022) to prepare stock in the new hub that has been running since January

Find the Most Proper Method

Determine the most appropriate and accurate forecasting approach to model weekly demand

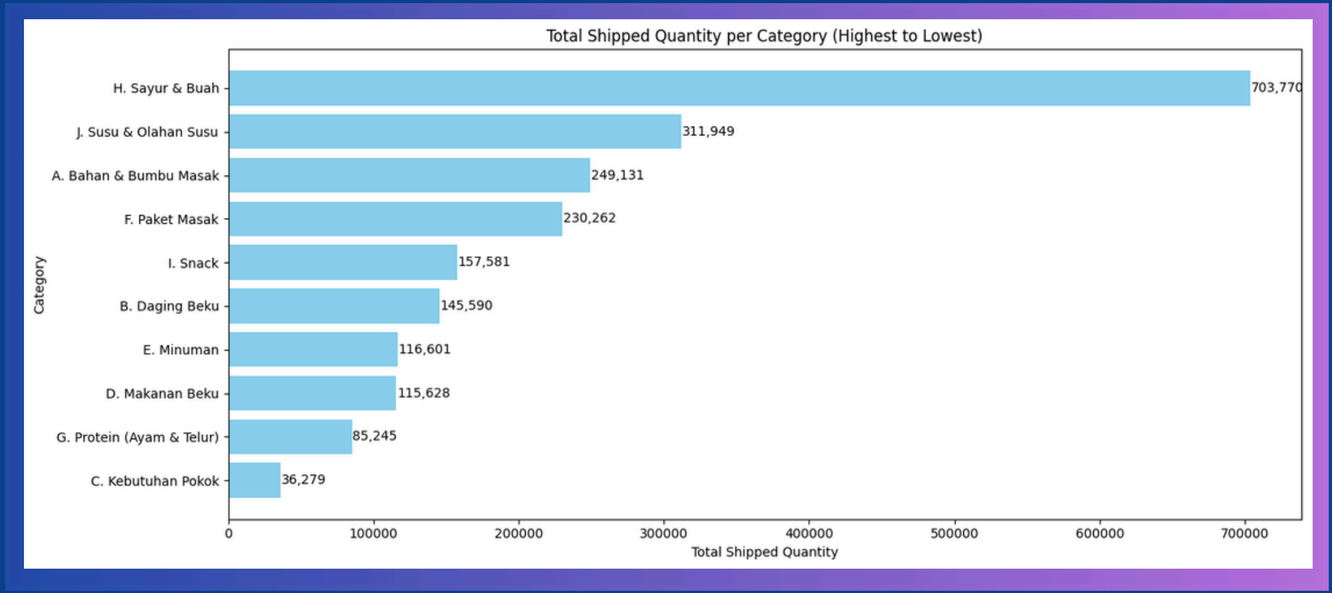
Forecast August 2022 Demand

Produce accurate weekly demand predictions for each hub across all 4 weeks of August 2022.

Insight & Recommendations

Translate forecast results into actionable stock operations strategy for the operation team.

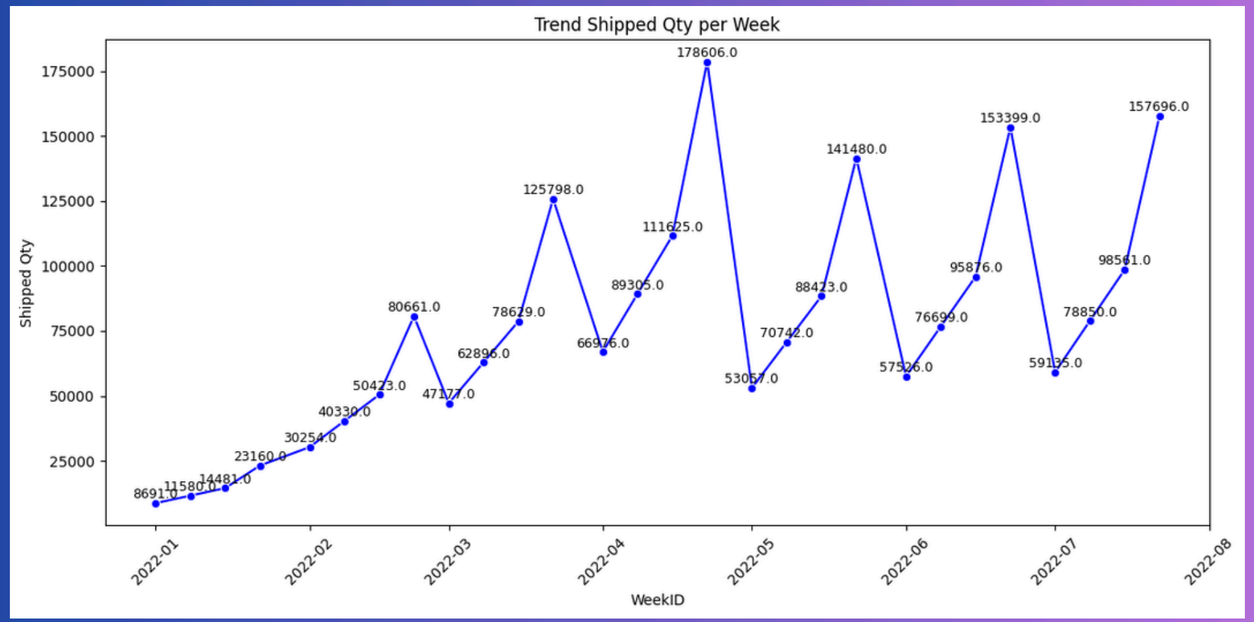
EDA – Total Shipped Quantity per Category



Key Insight

- **Fruits and vegetables** account for **32.7%** of the company's total shipments, with a **volume 2.26 times** larger than that of the second-ranked category (Milk & Dairy, 14.5%).
- **The top three categories** account for **58.8%** of the warehouse's operational volume.
- **Fruits and vegetables, and dairy products** account for **47.2%** of the total business and have the highest risk of spoilage (**waste**).
- **Low-volume categories** (essential goods) can be managed with **less frequent restocking.**

EDA – Trend Weekly & Time Series Decomposition



Outlier Check

The detection of 108 outliers using the IQR method found real volume, not data errors: 94 come from Vegetables & Fruits and Milk, and 55 fall in Week 4. All rows were kept.

Trend

Volume rose until the April Ramadan peak, fell 20.8% in May, then grew only 8.4% in June and 2.8% in July. Demand is levelling off.

Seasonal

Every month splits the same way: W1 15%, W2 20%, W3 25%, W4 40%. The peak is always Week 4, the 22nd to month end (7 to 10 days).

Pattern Type

The seasonal amplitude grows with volume while the weekly ratio stays fixed, so the pattern is multiplicative, not additive.

Why Focus on Vegetables & Fruits and Milk & Dairy Products?

Data-Driven Rationale

Focusing on these two categories accounts for **47.2%** of the company's total shipping volume **(1,015,719 units)**. Given that these two categories have the shortest shelf life, forecast accuracy is key to minimizing financial losses due to spoilage.

Vegetables & Fruits

Fruits and vegetables are among the most perishable food categories. According to industry literature, leafy greens can even start to wilt within a matter of days. Therefore, accurate weekly forecasts are crucial to ensuring that our inventory matches what will actually be sold, so we can minimize the buildup of leftover stock at the end of the week that risks going to waste.

Milk & Dairy Products

Although shelf life is slightly longer than vegetables (up to a week or slightly more under refrigeration), this product is highly sensitive to temperature fluctuations. Overstocking risks straining cold storage capacity, which is costly.

Source : https://agriculture.institute/food-fundamentals-cpo/shelf-life-of-different-food-categories/#google_vignette

Data Check: Clean Data with a Fixed Weekly Pattern

Data Quality

1,400 rows, exactly 7 months x 10 categories x 4 weeks x 5 hubs (January to July 2022). No missing values, no duplicates, and no zero or negative quantities. Only the data types needed fixing.

Fixed Weekly Split

In all 350 month x category x hub combinations, the month is split the same way: W1 15%, W2 20%, W3 25%, W4 40%. W1 to W4 start on the 1st, 8th, 15th, and 22nd, so W4 covers 7 to 10 days.

What It Means

Peak demand is always in W4, never W3. The weekly forecast is really one August total split 15/20/25/40, with only seven months of history, so simple methods can beat complex ones.

Modeling - Method Comparison & Selection

Moving Average

MAPE

30.6%

Underperformed

A flat average misses every Week 4 peak: 29.8% in June, 31.4% in July. SARIMA did worse (80.1% and 14.0%).

Holt-Winters (multiplicative)

MAPE

7.8% Overall

Strong Candidate

Fits the fixed weekly ratio: 5.3% in June, 10.4% in July. It still carries the early trend and puts August 10% above July.

Seasonal Naive

MAPE

5.2% Overall

Best Model

Repeats last month's same week. Lowest error where stock is planned: 9.2% across 50 hub x category series, 7.8% (June) and 2.7% (July) in total, each tested on a full unseen month.

Forecast Result – All Categories per Hub

Hub	August Total	W1	W2	W3	W4
A	66,211	9,931	13,243	16,553	26,484
B	68,077	10,210	13,616	17,020	27,231
C	85,879	12,882	17,177	21,470	34,350
D	85,378	12,806	17,075	21,345	34,152
E	88,697	13,306	17,739	22,173	35,479

Insight

- All hubs peak in Week 4 (August 22 to 31): 157,696 of the 394,242 units forecast for August.
- Hub E needs the most stock (88,697 units) and Hub A the least (66,211).
- Target stock is detailed for Vegetables & Fruits and Milk & Dairy, which carry 47.2% of volume and spoil fastest. The other 8 categories are in the notebook.

Forecast Result - Vegetables & Fruits

Hub	Safety Stock W4	W1	W2	W3	W4
A	529	2,557	3,411	4,263	6,820
B	423	2,046	2,728	3,411	5,456
C	882	4,263	5,684	7,104	11,367
D	1,147	5,541	7,388	9,236	14,778
E	1,235	5,968	7,957	9,946	15,914

Insight

- All hubs peak in Week 4 (August 22 to 31), which covers 10 days and 40% of the month.
- Safety stock is 8.4% of each week's forecast (95% service level, from June and July forecast errors). The column shows the W4 buffer; Hub E needs the most.
- The weekly figure shown is the Total Inventory Target (Forecast + Buffer), which can be used as a reference for Operation Team.

Forecast Result – Milk & Dairy Products

Hub	Safety Stock W4	W1	W2	W3	W4
A	414	1,384	1,846	2,308	3,692
B	345	1,153	1,538	1,922	3,077
C	689	2,308	3,077	3,845	6,152
D	758	2,538	3,384	4,230	6,767
E	804	2,692	3,589	4,487	7,178

Insight

- All hubs peak in Week 4 (August 22 to 31), which covers 10 days and 40% of the month.
- Safety stock is 12.6% of each week's forecast, thicker than Vegetables & Fruits, because milk grew two months in a row (June 6%, July 9%). The column shows the W4 buffer.
- The weekly figure shown is the Total Inventory Target (Forecast + Buffer), which can be used as a reference for Operation Team.

Stock Recommendation

Vegetables & Fruits and Milk - Priority

Restock Daily / Every 2-3 days

- Build up W4 stock (Aug 22 to 31) from the seasonal naive forecast
- Buffer 8.4% for vegetables, 12.6% for milk
- Real-time stock monitoring at all hubs
- Cold chain 0 to 4 °C for milk and leafy greens
- FIFO rotation

Semi Perishable

Restock Weekly

- Use last month's weekly pattern to guide orders
- Inventory Planning for Promotions & Holidays
- Moderate buffer acceptable
- Bahan & Bumbu, Paket Masak : restock weekly
- Protein & Daging Beku: BI-weekly review

Non-Perishable & Frozen

Bulk Monthly Restock

- Consolidate orders across hubs to save cost
- Larger safety stock acceptable
- Kebutuhan Pokok : monthly review (1.2% of July volume)
- Monthly stock review replaces weekly check

Limits & Next Steps

Limits

What the data cannot show

- Seven months of history, including the launch and Ramadan
- Error measured on only two test months, June and July
- Shipments only: no closing stock, waste, or stockouts
- Safety stock assumes a one-week lead time
- Independence Day (Aug 17) falls in W3; its effect is not measurable

Next Step

Correct orders after W1

- W1 is always 15% of the month
- W1 actual divided by 0.15 gives the August total
- Adjust W2 to W4 orders from that total
- Recompute buffers each month with new errors
- Rerun the notebook when a new month closes

Open Questions

Input needed from other teams

- Procurement: supplier lead time for fresh items
- Warehouse: cold storage room for W4 milk stock
- Marketing: Independence Day promotions in W3
- Operations: waste and stockout records per hub

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Thank You For Attention

Appendix